192.168.1.0/24 - Subnet the networks to accommodate the required hosts

Network 1

	Nearest Power	Usable IPs
Subnetting	2^7 = 128	.1 - .126

	Count	Total
Hosts:	65 Hosts + PC + 2 Switches	68

	Low End	High End
Range:	192.168.1.0	192.168.1.127

	Address	CIDR Notation
Subnet Mask:	255.255.255.128	/25
Broadcast:	192.168.1.127	

Network 2

	Nearest Power	Usable IPs
Subnetting	2^5 = 32	.129 - .158

	Count	Total
Hosts:	27 Hosts + PC + 2 Switches	30

	Low End	High End
Range:	192.168.1.128	192.168.1.159

	Address	CIDR Notation
Subnet Mask:	255.255.255.224	/27
Broadcast:	192.168.1.127	

Network 3	Nearest Power	Usable IPs
Subnetting	2^2 = 4	.161 - .162

	Count	Total
Hosts:	2 Routers	2

	Low End	High End
Range:	192.168.1.160	192.168.1.163

	Address	CIDR Notation
Subnet Mask:	255.255.255.252	/30
Broadcast:	192.168.1.163	

Configure all hosts and interfaces that require an IP address

Network 1

192.168.1.0/25

Subnet Mask	Default Gateway
255.255.255.128	192.168.1.1

Name	IP
Router 0 (GigabyteEthernet0/0)	192.168.1.1
Switch 0 (FastEthernet0/1)	192.168.1.2
Switch 1 (FastEthernet0/1)	192.168.1.3
PC0	192.168.1.4

Network 2

192.168.1.128/27

Subnet Mask	Default Gateway
255.255.255.225	192.168.1.129

Name	IP
Router 1 (GigabyteEthernet0/0)	192.168.1.129
Switch 2 (FastEthernet0/1)	192.168.1.130
Switch 3 (FastEthernet0/1)	192.168.1.131
PC1	192.168.1.132

Network 3

192.168.1.160/30

Subnet Mask	
255.255.255.225	
Name	**IP**
Router 0 (GigabyteEthernet0/1)	192.168.1.161
Router 1 (GigabyteEthernet0/1)	192.168.1.162

Using RIP ensure the routers can communicate with each other and route traffic to the appropriate networks

Router0 (R0)

```
R0(config)#router rip
R0(config-router)#version 2
R0(config-router)#network 192.168.1.0
R0(config-router)#network 192.168.1.160
R0(config-router)#exit
```

Router 1 (R1)

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.128
R1(config-router)#network 192.168.1.160
R1(config-router)#exit
```

Ensure that PC0 and PC1 can ping each other

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.132

Pinging 192.168.1.132 with 32 bytes of data:

Reply from 192.168.1.132: bytes=32 time<1ms TTL=126
Reply from 192.168.1.132: bytes=32 time<1ms TTL=126
Reply from 192.168.1.132: bytes=32 time<1ms TTL=126
Reply from 192.168.1.132: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.1.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=126
Reply from 192.168.1.4: bytes=32 time<1ms TTL=126
Reply from 192.168.1.4: bytes=32 time<1ms TTL=126
Reply from 192.168.1.4: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC0 to PC1

PC1 to PC0

Do a base configuration and rename the Routers and Switches:

Access

Password Type	Password
Enable Password	cisco
Privileged EXEC (secret)	class
Virtual Terminal	line
Console Line	letmein

CHECKLISTS

Router0 (R0)

Checklist	Complete
Hostname	R0
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.161 / 255.255.255.252
RIP	192.168.1.0 + 192.168.1.160
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

Router1 (R1)

Checklist	Complete
Hostname	R0
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.162 / 255.255.255.252
RIP	192.168.1.128 + 192.168.1.160
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

PC0

IP Address	Subnet Mask	Default Gateway
192.168.1.4	255.255.255.128	192.168.1.1

PC1

IP Address	Subnet Mask	Default Gateway
192.168.1.132	255.255.255.224	192.168.1.129

Switch0 (S0)

Checklist	Complete
Hostname	S0
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.2 / 255.255.255.128
Default Gateway	192.168.1.1
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

Switch1 (S1)

Checklist	Complete
Hostname	S1
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.3 / 255.255.255.128
Default Gateway	192.168.1.1
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

Switch2 (S2)

Checklist	Complete
Hostname	S2
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.130 / 255.255.255.224
Default Gateway	192.168.1.129
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

Switch3 (S3)

Checklist	Complete
Hostname	S3
Banner MOTD	"Any unauthorized access will not be tolerated!"
Date & Time	clock set
IP Address / Subnet Mask	192.168.1.131 / 255.255.255.224
Default Gateway	192.168.1.129
enable / secret / virtual / console / SSH	cisco / class / line / letmein / shell

What switches are Root Bridges?

Switch	Root Status	Priority
S0	Root Bridge (Network 1)	0
S1	Secondary Root Bridge (Network 1)	4096
S2	Root Bridge (Network 2)	0
S3	Secondary Root Bridge (Network 2)	32768

```
S0#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID     Priority     1
              Address      0001.4218.742B
              This bridge is the root
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID   Priority     1   (priority 0 sys-id-ext 1)
              Address      0001.4218.742B
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/24           Desg FWD 19        128.24   P2p
Fa0/23           Desg FWD 19        128.23   P2p
Gi0/1            Desg FWD 4         128.25   P2p

S1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID     Priority     4097
              Address      0005.5E85.2099
              This bridge is the root
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID   Priority     4097  (priority 4096 sys-id-ext 1)
              Address      0005.5E85.2099
              Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
              Aging Time   20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/23           Desg FWD 19        128.23   P2p
Fa0/24           Desg LSN 19        128.24   P2p
```

```
S2#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0002.170A.E9DA
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0002.170A.E9DA
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/1            Desg FWD 4         128.25   P2p
Fa0/23           Desg FWD 19        128.23   P2p
Fa0/24           Desg FWD 19        128.24   P2p
S3#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0002.170A.E9DA
             Cost        19
             Port        23(FastEthernet0/23)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00E0.F798.0C10
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/24           Altn BLK 19        128.24   P2p
Fa0/23           Root FWD 19        128.23   P2p
```